Gibson assembly of three fragments

Version 2.1 · Smith lab · Updated 12 March 2026

1. Overview

Gibson assembly joins overlapping DNA fragments in a single isothermal reaction at 50 °C. The 5′ exonuclease creates single-stranded overhangs, the polymerase fills the gaps and the ligase seals the nicks. See the NEB protocol for details.

2. Reagents

Component	Volume (µl)	Final concentration
Vector (linearised)	1.0	50 ng
Insert fragments	2.0	3:1 molar ratio to the vector backbone, adjusted by length
2× Gibson master mix	10.0	1×
Nuclease-free water	to 20	–

3. Procedure

1. Thaw the master mix on ice and spin it down briefly.
2. Combine the fragments with the master mix in a PCR tube on ice. Mix by pipetting up and down gently five times, then spin down.
    a. For 2–3 fragments, incubate 15 min.
    b. For 4–6 fragments, incubate 60 min.
3. Incubate at 50 °C in a thermocycler.
4. Transform 2 µl into 50 µl of competent cells (10^8 cfu/µg). Plate on LB with 100 µg/ml ampicillin and grow overnight at 37 °C in 5% CO_2-free incubator.

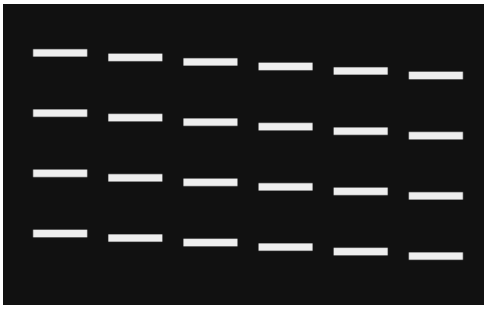

Figure 1. Colony PCR of four clones; expected band at 1.2 kb.

4. Troubleshooting

- No colonies: check the competent cells with a pUC19 control.
- Many empty vectors: DpnI-digest the PCR products, or gel-purify the vector.

This section is laid out in two columns to test column detection. The left column continues for a while with ordinary sentences about cloning, primers, overlaps and the assembly temperature, so that it wraps over several lines in the column.

Keep the overlaps between 20 and 40 bp with a melting temperature above 48 °C. Avoid secondary structure at the ends of the fragments.

The right column should be read after the left column, not interleaved with it line by line, which is what naive extraction would do.

Use `primer3 --overlap 25` to design the overlaps.